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Appendix A: A surface-code-based delayed choice network

For comparison, we design a surface-code-based delayed-choice quantum network. Any variable not explained here has been adopted from the main paper.

In this network, there is no teleportation involved between a separate storage code and an input code since we are dealing with surface codes directly. Hence, the storage-and-transport code is the same as the input/output code.

Let n_{ms} be the number of qubits in the surface code patch. Its distance will be $\sqrt{n_{ms}}$. Surface code LFR R_{ss} per cycle is given by:

$$R_{ss} = 0.3(70p_g)^{\frac{\sqrt{n_{ms}}+1}{2}} \tag{A1}$$

Bell Pair Creation: First, Charlie Inc. creates Bell-pairs. This involves $2\sqrt{n_{ms}}$ error correction cycles. Its LFR is given by:

$$R_{1s} = 2\sqrt{n_{ms}}R_{ss} \tag{A2}$$

The time taken for this is computed as:

$$T_{1s} = 6t_g\sqrt{n_{ms}} \tag{A3}$$

For every Bell-pair Charlie Inc. creates, he keeps the surface code patch containing one half of the Bell-pair to himself, while he transports the other half to an end-node.

Transport: During transport, the surface code undergoes $\frac{T_{3s}}{6t_g}$ cycles, where T_{3s} is the transport time. The LFR during this time is given by:

$$R_{3s} = \frac{T_{3s}}{6t_g}R_{ss} \tag{A4}$$

Measurement: The surface code patches are measured as soon as the truck arrives at the end-node qATM. The time taken to measure a surface code patch is $6t_g$, and the measurement error is R_{ss}. Since each surface code has just one qubit, we can measure each of these surface codes together in parallel. Since our required bit-rate is r_e, we can measure r_e number of surface codes every second in parallel to obtain the required bandwidth.

Now, we have the current net LFR after distribution:

$$R_{nets} = R_{1s} + R_{3s} + R_{ss} \tag{A5}$$

And the total time taken for this is:

$$T_{tots} = T_{1s} + T_{3s} + 6t_g \tag{A6}$$

Storage and Measurement: Meanwhile, Charlie Inc., on his side, needs to store his halves of the Bell-pairs, at least until the other halves are unloaded i.e. until T_{tots} amount of time has passed. The LFR for this is given by:

$$R_{stores} = 2\frac{T_{tots}}{6t_g}R_{ss} \tag{A7}$$

The factor 2 comes from the fact that Charlie Inc. is storing two halves, each corresponding to that of Alice and Bob respectively.

Next, Charlie Inc. performs a Bell-measurement between the two surface code patches. The LFR for this is given by:

$$R_{BMs} = 4\sqrt{n_{ms}}R_{ss} \tag{A8}$$

Overall: The total LFR for the whole protocol is:

$$R_{tots} = R_{BMs} + R_{stores} + 2R_{nets} \tag{A9}$$

Logistics: Since each logical qubit can be measured in parallel, the number of patches we need per second is equal to the desired bandwidth r_e. Considering S number of end-nodes and trucks or train cars with capacity of n_t physical qubits, the number of trucks per second is given as $N_{trucks} = \frac{Sr_e}{\left(\frac{n_t}{n_{ms}}\right)}$. The life cycle of a truck is the total time T_{tots} from loading to unloading plus the time T_{3s} needed to go back. Therefore, the total number of trucks required for the network is given by:

$$N_{truck_{tots}} = \frac{Sr_e}{\left(\frac{n_t}{n_{ms}}\right)}(T_{tots} + T_{3s}) \tag{A10}$$

Algorithm 2 Secure Delayed Choice Quantum Network Protocol With Photonic Measurements

```
for all end-users do
    Charlie Inc. prepares Bell pairs encoded in surface codes
    Charlie Inc. loads each half of the Bell pairs into separate sets of qLDPC code blocks via teleportation
    Charlie Inc. stores one set of qLDPC blocks and transports the other set to destination
    End-user puts in a purchase order for n_bits number of bits as per their requirement
    for i from 1 to n_bits do
        qATM at the destination unloads a logical qubit from a qLDPC block onto a surface code
        qATM converts surface code physical qubits to photons in a pre-defined order via pulsed excitation and de-excitation
        Photons sequentially sent to end-user's smartphone and measured
        Photonic measurements determine the logical surface code measurement, leading to a resultant measured bit
        Bit stored in the end-user's smartphone for later use.
    end for
end for
Publish measurement bases
for users Alice and Bob wishing to connect do
    Alice and Bob decide to correlate n_key number of bits
    for i from 1 to n_key do
        Charlie Inc. unloads two corresponding qLDPC qubits onto two surface code patches
        Charlie Inc. performs Bell measurement on surface code patches
        Charlie Inc. broadcasts the measurement result to Alice and Bob, resulting in a correlated key bit
    end for
    Alice and Bob perform standard reconciliation techniques and get the final secure key
end for
```